PS02 Rice Leaf Disease Classification

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GitHub : https://github.com/Sundaranandhan15/DL-CBL-PS02/

1. Problem and Objective

Rice leaf diseases reduce yield, and early lesions of different diseases look alike to non-experts, which delays correct treatment. The task is to classify a rice leaf photo into one of four classes: bacterial blight, blast, brown spot or tungro. Success is measured on a held-out test set by macro F1 (equal weight to every class) together with per-class recall, because overall accuracy can hide a weak class.

2. Dataset and Preprocessing

The Rice Leaf Disease Image Samples dataset (Sethy et al., Mendeley Data, doi:10.17632/fwcj7stb8r) was used in full: 5932 images (bacterial blight 1584, blast 1440, brown spot 1600, tungro 1308). A stratified random split with seed 42 gave 4152 train, 890 validation and 890 test images (70/15/15; test = 238/216/240/196). The classes are fairly balanced (22–27% each), so no class weights or resampling were used. Images were resized to 224×224 and scaled to [−1, 1] with MobileNetV2's *preprocess_input*. Augmentation (horizontal flip, rotation 0.08, zoom 0.10, contrast 0.10) sits inside the model and is active only in training, so validation and test images are never augmented.

3. Methodology

A pretrained MobileNetV2 (ImageNet weights, top removed) was used as a fully frozen feature extractor, followed by global average pooling, dropout 0.30 and a 4-way softmax layer. Only the 5,124 parameters of the final layer were trained (2.26 M in total). The checkpoint with the lowest validation loss was kept, and the learning rate was reduced when validation loss stalled. Table 1 lists the hyperparameters; results are in Table 2.

Item	Setting	Item	Setting
Backbone	MobileNetV2, frozen	Loss	Sparse categorical cross-entropy
Optimiser	Adam, lr 1e-3	LR schedule	ReduceLROnPlateau (×0.2, patience 2, min 1e-7)
Batch size	32	Epochs	Max 10 (all 10 ran)
Early stopping	val_loss, patience 4, best weights	Input / seed	224×224 / 42 (TensorFlow 2.21.0, GPU)

Table 1. Hyperparameters.

4. Results

Model	Test acc. (%)	Macro F1 (%)	F1 BB / Blast / BS / Tungro (%)	Recall Blast (%)	Train time
MobileNetV2 (frozen)	96.52	96.56	95.65 / 92.64 / 97.93 / 100.00	90.28	≈2.1 min

Table 2. Test-set results (n = 890). BB = bacterial blight, BS = brown spot. Test loss 0.10; training time is the sum of per-epoch times on a Colab GPU.

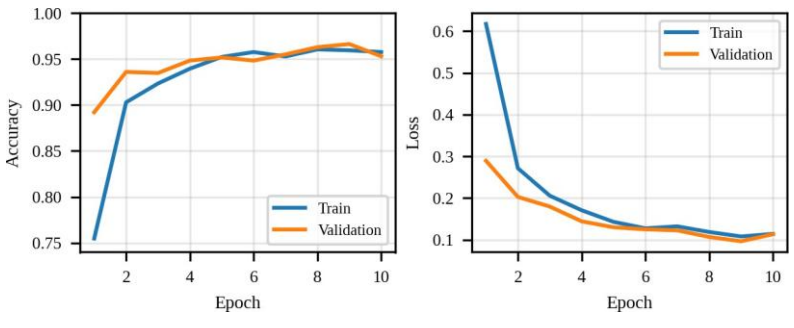


Fig. 1. Training and validation accuracy and loss per epoch (validation split).

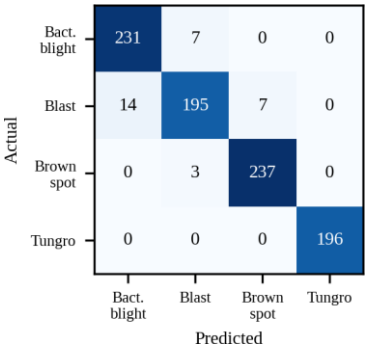


Fig. 2. Confusion matrix on the test set.

5. Analysis and Discussion

Of 890 test images, 31 were misclassified. Tungro was perfect (196/196), while blast was the weakest class: 14 blast leaves were labelled bacterial blight and 7 brown spot, and 7 bacterial blight leaves were labelled blast (Fig. 2). The misclassified images (viewed in the notebook) mostly show small or early lesions, blurred or cluttered backgrounds, and leaves with yellow-brown streaks that resemble bacterial blight, so lesion colour and shape overlap; four of the eight errors shown had confidence below 55%.

Training and validation curves stay close (final training accuracy 95.76%, validation 95.30%), so there is no over-fitting. Validation loss was still falling until epoch 9 and early stopping never triggered, which suggests mild under-fitting from the frozen backbone. The challenge of variable field images was handled with ImageNet transfer and augmentation; no ablation was run, so their individual effect is not measured.

6. Real-world Relevance, Limitations and Ethics

The model could drive a smartphone advisory tool that helps farmers and extension workers pre-screen leaves. It is not ready for deployment: all images come from one dataset, the split is random per image so near-duplicate photos could sit in both train and test (not checked), and robustness to other fields, cameras, growth stages and healthy leaves is untested (there is no healthy class). A wrong label could cause needless or missed pesticide use, so outputs should be shown as suggestions with confidence and confirmed by an expert. Field photos may also reveal farm location.

7. Conclusion and Future Work

A frozen MobileNetV2 reached 96.52% test accuracy and 96.56% macro F1 on the 890-image test split, with blast the hardest class. Next I would fine-tune the top backbone layers and add Grad-CAM to check that predictions rely on lesions rather than background.

Dataset: Sethy, P. K., Barpanda, N. K., Rath, A. K. and Behera, S. K. (2020), Rice Leaf Disease Image Samples, Mendeley Data, doi:10.17632/fwcj7stb8r; paper: Computers and Electronics in Agriculture 175, 105527.